

Oral Examination – Student Handout

Software Construction 2 · Tasks 1-4 and Bonus

Task 1. Core OOP concepts, warm-up

Explain each concept shortly and give one Java example where helpful. Please talk through your thoughts aloud.

- class vs. object
- encapsulation and why fields should usually be private
- inheritance vs. composition (is-a vs. has-a)
- method overloading vs. overriding

Task 2. True/False questionnaire: OOP and Java basics

Answer each statement with true or false. **Please talk through your thoughts aloud while answering; explain why you choose true or false.**

Statement	True	False
In Java, a class can extend only one other class.		
A private field can be accessed directly from any other class in the same package.		
A constructor has the same name as the class and no return type.		
Method overloading means same method name with different parameter lists.		
A subclass can override an inherited method by providing a compatible method with the same signature.		
An interface can be implemented by multiple unrelated classes.		
A static field belongs separately to each object instance.		
For objects, == compares references, not logical content.		

Task 3. 2-minute writing task: small method

Write a Java method `countPositive(int[] values)` that returns the number of values greater than 0. Then explain what should happen for `null` and for an empty array.

Task 4. Debugging task: find and repair errors

Mark at least four problems in the code and explain how you would correct them.

```
1  public class BankAccount {
2      public double balance;
3
4      public void BankAccount(double balance) {
5          balance = balance;
6      }
7
8      public deposit(double amount) {
9          balance += amount;
10     }
11
12     public void withdraw(double amount) {
13         balance - amount;
14     }
15 }
```

 Java

Bonus. Code reading: dynamic dispatch

Predict the output and explain why. Then answer whether `Circle c = new Shape();` compiles.

```
1  class Shape {
2      public String name() { return "shape"; }
3      public void print() { System.out.println(name()); }
4  }
5
6  class Circle extends Shape {
7      @Override
8      public String name() { return "circle"; }
9  }
10
11 public class Demo {
12     public static void main(String[] args) {
13         Shape s = new Circle();
14         s.print();
15     }
16 }
```

 Java

Notes and answers



Additional notes and answers

